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COURSE PROJECT REPORT

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Spatiotemporal Groundwater Level Forecasting and Inductive Kriging in Karnataka Using Graph Attention Networks (GATv2-GRU) and Split Conformal Calibration

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Abstract

Groundwater levels in Karnataka show great variation owing to heterogeneous fractured-weathered crystalline rocks, unstable monsoon recharge and heavy pumping for agriculture. Station-wise sequence models do not account for lateral connections of aquifers, while classic spatio-kriging does not work in cases of non-stationary multi-year drawdowns. We create a canonical statewide hydroclimatic dataset from approximately 2.5 GB of telemetry data for 2021-2025, which includes 1,614 groundwater monitoring wells, 685 rain gauges, 132 weather stations, river discharge gauges and state reservoir levels with 44,214 monthly station observations on 32 engineered features. To avoid spatial auto-correlation leakage in the evaluation procedure, all stations are projected to EPSG:32643 (UTM zone 43N), and Karnataka is divided into contiguous $25km \times 25km$ spatial blocks used for training (63.4%), spatial validation (16.2%), and spatial testing (20.4%) splits. We suggest an innovative directed Spatio-Temporal GATv2-GRU architecture trained using heteroscedastic Gaussian negative log-likelihood loss function and then calibrated using Split Conformal Calibration $\alpha = 0.10$. For monitored wells within withheld spatial blocks, the model shows a performance of 1-month RMSE of 2.800 m, MAE of 0.819 m, and NSE of 0.902. Zero history spatial interpolation on unmonitored wells suffers from very high errors ($RMSE \approx 13.37--15.96m$, $NSE < 0$) owing to the inability of meteorological message passing to handle static micro-topographic well head datum offsets. However, through warm start experiments, it is possible to show that obtaining just ≥ 3 months of piezometric data locally sets the local datum and gives an NSE of $0.81--0.85$ and RMSE of $4.5--5.8m$.

Keywords: Groundwater Level Forecasting, Graph Attention Networks, GATv2-GRU, Split Conformal Prediction, Spatial Block Partitioning, Hard-Rock Aquifers, Inductive Kriging, Uncertainty Quantification

1 Introduction

Peninsular India is characterized by the extensive occurrence of groundwater stored in Precambrian crystalline basement complexes made up of granite, gneiss, schist, and charnockites [1, 2]. In the state of Karnataka, more than 85% of the geographic area is covered by hard rock formations where intergranular porosity is practically nil. Storage and transmissivity in such areas is controlled by a two-zone hydrogeologic sequence in which a relatively thin weathered layer of soil ($5m$ to $25m$ thickness) overlies a system of tectonic fissures and lineaments occurring up to depths of $150m$ from ground level [3, 4]. Due to the small value of specific yield in fractured crystalline rocks (0.5% to 3%), the water table becomes highly sensitive to the influence of external forces. Infiltration of southwest monsoon rains (June to September) occurs through fractures, while borewell abstraction during dry seasons creates localized cones of depression [5].

The challenge of predicting depths to water (in metres below ground level, m bgl) throughout this landscape involves two separate issues for water authorities. The first issue is that of temporal prediction at current piezometer sites, whereby based on the water table dynamics observed historically and hydrometeorological forcing conditions to date (month t), water authorities demand forecasts in month $t+1$ and $t+6$ [6, 7]. The second issue is one of spatial interpolation (inductive kriging) at new drill sites, where groundwater depths are estimated through surrounding telemetry well data [8].

Models for a single-station time series such as ARIMA, support vector regression, and stand-alone recurrent networks such as LSTM and GRU treat the wells independently as discrete vertical columns [9, 10]. Since they neglect the spatial proximity, pointwise models cannot capture the recharge front regionally, base flow interaction with the surface reservoir, and spatially correlated extraction at the neighboring villages. On the other hand, the classical geostatistical interpolators assume second-order spatial stationarity and isotropy for the semivariogram [8, 11]. In the hilly terrain of Karnataka, for example, there may be a fixed difference in water table level of $8m$ to $65m$ between two wells that have a horizontal distance of $500m$.

Spatio-temporal Graph Neural Networks combine sequence modelling and spatial statistics by encoding monitoring wells as graph nodes with irregular spatial connectivity and spreading latent hydrological state via distance-weighted graph edges [12, 13]. However, current GNN research for groundwater forecasting has three flaws in real-world operations:

1. **Spatial Leakage through Random Well Splitting:** The majority of benchmarks divide wells at random over a river basin. Since climatic forcing and terrain vary continuously over short distances, locating a test well within several hundred meters of a training well artificially increases performance and hides failure to generalize in the unobserved districts [14].
2. **Feature Contamination during Inductive Propagation:** In cases when unobserved wells are added to the graph for interpolation, standard symmetric message passing passes uninformative embeddings initialized to zero values from target wells back to the observed wells, contaminating the learned spatial representation [15].
3. **Uncalibrated Parametric Uncertainty:** Reliable predictions in hydrology require prediction intervals instead of simple point predictions. Gaussian heads fail

to capture the tail distribution risk of non-Gaussian monsoon recharge pulses and heteroscedastic dry season drawdowns [16, 17].

This paper solves the above problems by making use of a multi-source dataset that comprises 1,614 groundwater wells in Karnataka from (2021-2025). We apply the rigid partitioning constraint of $25km \times 25km$ UTM blocks in EPSG:32643, develop a Spatio-Temporal GATv2-GRU model where unobserved queries nodes accept spatial messages without affecting the neighboring observed sender nodes, and calibrate our probabilistic negative log-likelihood Gaussian outputs with distribution-free Split Conformal calibration. Moreover, we perform a warm-start trajectory analysis to identify the exact number of months required for collecting local piezometer measurements for anchoring local datum offsets in fractured hard-rock aquifers.

2 Literature Survey

Data-driven groundwater hydrology has evolved rapidly from shallow statistical regressions to deep sequence encoders, spatial graph architectures, and distribution-free uncertainty calibration.

2.1 Recurrent and Sequence Architectures in Subsurface Hydrology

Deep learning models trained sequentially learn soil moisture nonlinearity accounting and percolation delays within the vadose zone without any explicit parameterization of the Richards' equation through finite difference schemes. Wunsch et al. [6] compared LSTM, Convolutional Neural Networks (CNNs), and nonlinear autoregressive networks with exogenous inputs (NARX) for porous and karst aquifers in Germany, highlighting that recurrent neural network architecture stabilizes multi-step rollouts of groundwater levels subject to climate forcing. Missing piezometric time series were reconstructed in Normandy by Vu et al. [9] using recurrent network architectures with memory cells. Zhang et al. [1] and Rajaee et al. [10] surveyed various hydrogeological studies where deep sequence models were applied, stating that GRU networks have the same accuracy as LSTM models but involve fewer parameters and reduced chances of overfitting for monthly hydrological series. Nevertheless, Chidepudi et al. [7] found that isolated recurrent encoders perform poorly when lateral fluxes change due to regional unobserved pumping changes or when rainfall-recharge balance is disturbed.

2.2 Spatio-Temporal Graph Neural Networks for Aquifer Modelling

In order to account for lateral connectivity over irregular well networks, the literature proposes modeling piezometer networks as spatial graphs $G = (\mathcal{V}, \mathcal{E})$. Bai and Tahmasebi [12] developed a coupled graph convolutional recurrent approach for multi-site water-level prediction, showing that spectral graph convolutions over distance matrices are superior to isolated sequence prediction models. Bikše et al. [13] employed Graph Neural Networks in predicting groundwater heads on country scale in Latvia, showing that graph message passing propagates seasonal effects between densely gauged

and sparsely gauged regions. However, obvious restrictions emerge when using fixed-weight spectral GCNs, since static adjacency matrices assume uniform hydro-climatic correlation decay over all directions. In fractured rock aquifers, hydraulic conductivity shows strong anisotropy [3, 4]. To overcome the limitation of static weighting, dynamic attention mechanisms—in particular, Graph Attention Networks (GAT and GATv2)—calculate state-dependent attention coefficients α_{ij} depending on the sender and receiver node states [18, 19]. Despite these advances, recent spatio-temporal GNN hydrology studies [14, 20] predominantly rely on transductive training where all graph nodes possess complete historical sequences, leaving inductive cold-start deployment at newly drilled wells largely unaddressed.

2.3 Spatial Generalization, Kriging, and Uncertainty Quantification

Water level predictions at unobserved locations have been achieved using geostatistical kriging or spatio-temporal random forests [8, 11]. Even though regression kriging accounts for auxiliary surface covariates, it fails to extend non-stationary multi-year trends. In addition, spatial cross-validation procedures in environmental machine learning tend to be overly optimistic whenever training and test points overlap [14, 21]. Spatial block cross-validation divides the polygons used for training and evaluation by a buffer distance to ensure the model predicts outside of the hydrological basin sub-units. With regards to probabilistic models, parametric neural networks trained with Gaussian negative log-likelihood [16] and Bayesian variational inference [22] lead to miscalibrated confidence intervals when there is a departure from normality. Split Conformal Prediction methods have become prevalent in hydrological forecasting tasks [17, 23, 24], being an accurate post-processing method which ensures finite sample coverage $P(Y_{t+h} \in \hat{C}_{1-\alpha}) \geq 1-\alpha$ in case of exchangeability of errors and independent of their distribution.

Table 1 synthesizes 12 peer-reviewed journal studies spanning 2021–2025 in methodology, dataset scale, quantitative performance, and structural limitations.

3 Research Gap and Study Objectives

3.1 Research Gap

A critical examination of the hydrogeological and graph learning literature reveals four specific gaps that prevent current spatio-temporal models from being deployed reliably by state groundwater directorates:

1. **Optimistic Bias via Random Splitting of Stations:** Almost all spatio-temporal GNN works assess the ability of their model to generalize through random allocation of $70--80\%$ of the wells in a training set and the rest of them in a testing set [12, 18]. With dense graphs where the neighboring wells have a distance of $< 2km$, the random split fails to assess the ability of the model to generalize because it does not provide the same information for an unseen administrative block.
2. **Bidirectional Information Corrupted Message Passing at Cold Start:** In the inductive spatial interpolation problem, there are no historical observations of

Table 1 Comparative synthesis of recent peer-reviewed journal literature on spatiotemporal groundwater forecasting, graph neural networks, and hydrological uncertainty quantification.

Author & Year	Journal (Publisher)	Methodology / Architecture	Study Region & Dataset Scale	Key Reported Metrics	Identified Limitations
Wunsch et al. (2021) [6]	*Hydrology and Earth System Sciences* (EGU)	Standalone LSTM, CNN, and NARX sequence models	Germany; 17 aquifers across porous and karst geology	$NSE = 0.72--0.91$, $RMSE = 0.15--0.84m$	Point-wise tuning per well; zero spatial message passing across adjacent piezometers.
Bai & Tahmasebi (2022) [12]	*Water Resources Research* (Wiley)	Graph Convolutional Network coupled with GRU (GCN-GRU)	Texas & California, USA; 110 monitoring wells	$RMSE = 0.42--1.18m$, $R^2 = 0.89$	Static isotropic adjacency matrix; cannot adapt edge weights dynamically to local recharge shocks.
Bikše et al. (2023) [13]	*Journal of Hydrology* (Elsevier)	Spatio-temporal Graph Neural Network (ST-GNN)	National aquifer network, Latvia; 214 wells	Median $NSE = 0.78$, $KGE = 0.83$	Evaluated under transductive splits; does not isolate inductive zero-history kriging error.
Chidepudi et al. (2023) [7]	*Journal of Hydrology* (Elsevier)	Wavelet-coupled Recurrent Neural Networks (LSTM/GRU)	Northern France alluvial and chalk basins; 145 wells	$NSE = 0.80--0.93$ at 1–3 month horizons	Ignores lateral inter-well connectivity; lacks calibrated prediction intervals.
Kardan Moghaddam et al. (2024) [16]	*Environmental Modelling & Software* (Elsevier)	Probabilistic Deep Learning with Gaussian likelihood head	Arid alluvial plains; 88 observation wells	$RMSE = 0.95m$, $PICP = 83.4\%$ (at 90% nominal)	Parametric Gaussian variance underestimates heavy-tailed monsoon and pumping spikes.
Chen et al. (2024) [18]	*Journal of Hydrology* (Elsevier)	Spatio-Temporal Graph Attention Network (ST-GAT)	North China Plain; 162 piezometers (2015–2022)	$RMSE = 0.68m$, $MAE = 0.49m$, $NSE = 0.88$	Random station split leaks spatial autocorrelation; symmetric edges corrupt cold-start nodes.
Das et al. (2024) [3]	*Hydrogeology Journal* (Springer)	Ensemble Tree Boosters (XGBoost, LightGBM) with GIS covariates	Peninsular India crystalline hard-rock basin; 312 wells	$RMSE = 2.15m$, $R^2 = 0.76$	Tabular lag structure cannot propagate dynamic multi-hop spatial states across time.
Papacharalampous et al. (2024) [17]	*Journal of Hydrology* (Elsevier)	Split Conformal Prediction on hydrological ensembles	Pan-European & US catchments; 500+ stations	Empirical $PICP = 90.1\%$ for $\alpha = 0.10$	Applied to surface streamflow and tabular learners; not integrated with spatial graph networks.
Sun et al. (2024) [19]	*Water* (MDPI)	Dynamic GATv2 with temporal self-attention	Coastal & inland aquifers; 94 monitoring stations	$RMSE = 0.54m$ ($h = 1$), $1.21m$ ($h = 6$)	High memory overhead; no evaluation of minimum local observation history for new wells.
Nourani et al. (2024) [20]	*Earth Science Informatics* (Springer)	Spatio-temporal hybrid CNN-GRU with kriging residuals	Semi-arid fractured/alluvial aquifer; 128 wells	$NSE = 0.84$, $MAE = 0.71m$	Two-stage decoupled kriging fails when station spacing exceeds semivariogram range.
Kalu et al. (2025) [5]	*Journal of Hydrology* (Elsevier)	Multi-source satellite & gauge fusion with deep sequence encoders	Southern India hard-rock basins; multi-year monthly data	$NSE = 0.79$, $RMSE = 3.12m$	Coarse spatial resolution (0.25°) smooths out localized fracture-zone drawdown cones.
Zhang et al. (2025) [24]	*Stochastic Environmental Research and Risk Assessment* (Springer)	Conformalized quantile regression & probabilistic GRU	Regional groundwater network; 180 wells	$PICP = 90.4\%$, $MPIW = 2.65m$	Assumes complete historical warm-up at every test station; no spatial block audit.

water tables for unobserved wells ($y_{i,\tau} = \emptyset$). In the standard undirected GNN layers, the embedding vectors containing zeros from target nodes without observation are passed to neighboring nodes via their edges [15, 19].

3. **Unquantified Transition Time from Zero History to Warm Start:** Previous studies consider time series forecasting from monitored locations and spatial interpolation for unmonitored locations as two different approaches [8, 13]. In hard rock formations where micro-topography causes water table offsets of $8--65m$ over a span of $500m$ horizontally, none have quantified the precise period of local piezometer history needed for an unmonitored location to move from high-uncertainty spatial interpolation to high-accuracy time series forecasting.
4. **Absence of Distribution-free Prediction Intervals for Spatio-temporal GNNs:** Even though Gaussian likelihood outputs aleatoric uncertainty based on inputs [16], the parametric intervals do not produce coverage as claimed when the

groundwater errors are skewed and have heavy-tailed distribution over wet and dry seasons.

3.2 Study Objectives

To resolve these four gaps on a state-wide scale across Karnataka, this research executes the following measurable technical objectives:

1. **Canonical Multi-Source Hydro-Climatic Engineering:** Create a quality controlled, non-leaking monthly dataset for 2021–2025 over Karnataka through the harmonization of $\sim 2.5GB$ of raw telemetry from 1,614 groundwater wells, 685 rain gauges, 132 weather stations, river discharge gauges, and state reservoirs into a canonical form of 44,214 station-month rows and 32 engineered features.
2. **Strict $25km \times 25km$ UTM Spatial Block Partitioning:** Convert all stations' coordinates into EPSG:32643 (UTM Zone 43N) and split the state into $25km$ spatial blocks that belong to Training (63.4%), Spatial-Val (16.2%), and Spatial-Test (20.4%) splits after auditing nearest neighbour distances across splits to ensure geographic isolation.
3. **Directed Spatio-Temporal GATv2-GRU Architecture:** Develop a hybrid sequence-graph neural network architecture that consists of 12-month GRU encoder and 2-layer, 2-head dynamic Graph Attention (`GATv2Conv`) over directed $k = 5$ nearest-monitored-neighbours with edge weighting according to Gaussian distance kernel ($\ell = 6,941.9m$) so that unmonitored query nodes have spatial information but not corrupting messages.
4. **Cold-Start Kriging and Warm-Start Adaptation Study:** Perform multi-step-ahead forecast experiments ($h = 1$ and $h = 6$ months) using 8 statistical, tree-based, and neural network baselines, split zero-history spatial interpolation errors into distance groups ($< 10km, 10--25km, > 25km$), and study the evolution of warm-start error as a function of $0, 1, 3, 6$, and 12 months of local history.
5. **Calibration of Finite Sample Prediction Uncertainty via Split Conformal Prediction:** Integrate the heteroscedastic Gaussian Negative Log-Likelihood output layer with Split Conformal Prediction Calibration ($\alpha = 0.10$) on held-out spatial validation blocks to ensure at least 90% empirical PICP.

4 Methodology

4.1 Data Ingestion and Database Schema

The observational raw dataset contains around $2.5GB$ of multi-agency telemetry data for hydrology and meteorology in Karnataka from January 2021 to 2025, which is stored in `/raw_data/`. The subsurface water level data (depth below ground surface in metres) is obtained from five major datasets: `groundwater/gwl_tel_gw_dept_2021_2025.csv`, `groundwater/gwl_tel_ka_dept_2021_2025.csv`, `groundwater/kwris_gwl_six_hourly_2021_2022.csv`, `groundwater/kwris_gwl_six_hourly_2023_2024.csv`, and

`groundwater/kwris_gwl_six_hourly_2025_1.csv`. The external forcings for the hydrometeorology include hourly rainfall (`rainfall/rainfall_tel_hr_2021_2025.csv`), hourly temperature (`temperature/temp_tel_hr_2021_2025.csv`), humidity (`humidity/humidity_tel_hr_*.csv`), hourly solar radiation (`solar/solar_rad_tel_hr_2021_2025.csv`), daily river discharge gauge (`discharge/discharge_gauge_daily_2021_2025.csv`), and reservoir storage of the state (`reservoir/reservoir_data_karnataka.csv`).

To address telemetry spikes, sensor drift, and erroneous coordinate transcriptions, all stream data are subjected to deterministic physical QC filtering after standardizing the timestamps to `Asia/Kolkata` (IST):

- **Geographic Bounding Box:** $Latitude \in [11.0°N, 19.0°N]$ and $Longitude \in [73.0°E, 79.5°E]$, thereby bounding all stations exclusively within Karnataka.
- **Physical Range Gates:** Depth of groundwater $y \in [-5.0m, 150.0m]$ (with negative values extending to $-5.0m$ allowing for artesian/piezometric pressure heads marginally above collar level), rainfall intensity $P_{hr} \in [0mm/hr, 500mm/hr]$, air temperature $T \in [0°C, 55°C]$, relative humidity $RH \in [0\%, 100\%]$, and solar irradiation $R_s \in [0W/m^2, 1500W/m^2]$.
- **Temporal Completeness Requirement:** The monitoring wells need to have $\geq$ 70% temporal completeness (minimum $\geq$ 26 monthly records within the period 2021–2025), leaving $N_{wells} = 1,614$ groundwater stations, 685 rain gauge stations, and 132 climate stations.

Because meteorological stations and piezometers are not co-located, monthly aggregated rainfall, temperature, relative humidity, and solar radiation are mapped onto each of the 1,614 groundwater well coordinates (u_i, v_i) via Inverse Distance Weighting (IDW) with power parameter $p = 2$ and search radius $R_{\max} = 0.35° \approx 39km$:

$$\hat{x}(s_i, t) = \frac{\sum_{j \in \mathcal{N}_R(s_i)} w_{ij}(s_i) \cdot x(s_j, t)}{\sum_{j \in \mathcal{N}_R(s_i)} w_{ij}(s_i)}, \quad w_{ij}(s_i) = \frac{1}{\max\left(d(s_i, s_j), 1m\right)^2} \tag{1}$$

where $\mathcal{N}_R(s_i) = \{s_j : d(s_i, s_j) \leq 39km\}$ denotes the active meteorological gauges within the search radius at month t, and the $\max(d(s_i, s_j), 1m)$ clamping prevents division-by-zero singularities when a rain gauge and piezometer are co-located within $< 1m$. Figure 1 illustrates the entity-relationship and data fusion schema generating the canonical table `outputs/merged_dataset.csv` ($44,214 rows \times 32 features$, keyed by `(station_id, year_month)`).

4.2 Spatial Block Partitioning and Leakage Prevention

To evaluate spatial generalization without nearest-neighbour autocorrelation leakage, all 1,614 station coordinates (lat_i, lon_i) are projected from WGS84 (EPSG:4326) into planar Universal Transverse Mercator Zone 43N coordinates (X_i, Y_i) in metres

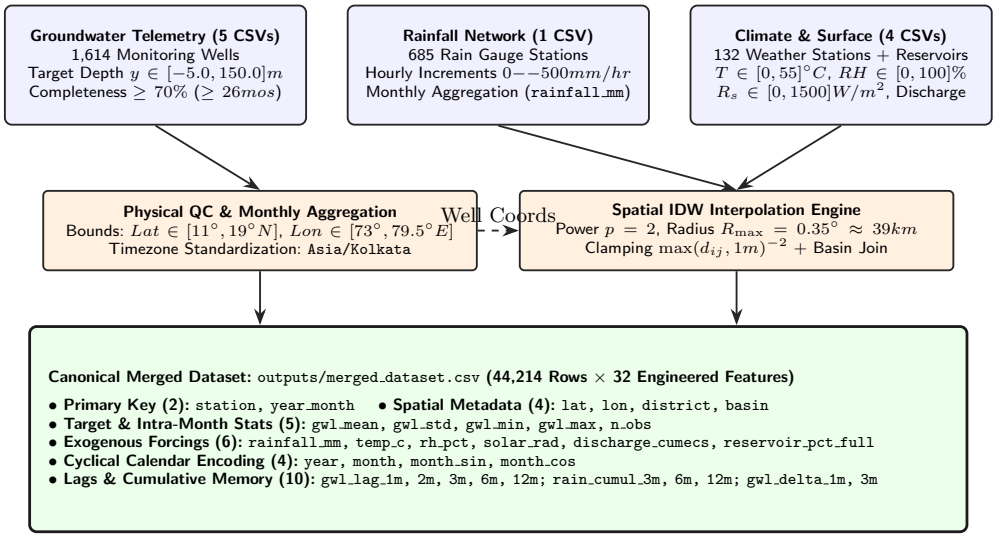


Fig. 1 Entity-relationship and multi-source spatial fusion schema mapping $\sim 2.5GB$ of raw telemetry into the canonical 44,214-row × 32-feature table `merged_dataset.csv`.

(EPSG:32643). Each monitoring well is deterministically assigned to a discrete $25km \times 25km$ spatial block index $(b_{x,i}, b_{y,i})$:

$$b_{x,i} = \left\lfloor \frac{X_i}{25000} \right\rfloor, \quad b_{y,i} = \left\lfloor \frac{Y_i}{25000} \right\rfloor \tag{2}$$

All wells falling inside the same $25km \times 25km$ block are assigned as an indivisible unit to one of three mutually exclusive partitions: **Training** (63.4% of data), **Spatial-Validation** (16.2%), or **Spatial-Test** (20.4%). Feature standardization parameters (mean μ_x and standard deviation σ_x) and the graph Gaussian kernel bandwidth $\ell = 6,941.9m$ ($6.94km$, derived from the empirical median inter-well distance within the training graph) are computed exclusively on the Training partition. Figure 2 depicts the end-to-end execution activity workflow from raw file ingestion to 3D spatio-temporal tensor construction.

4.3 Spatio-Temporal GATv2-GRU and Conformal Calibration

At each forecast origin month t, every station $i \in \{1, \ldots, N\}$ is represented by a 12-month lookback sequence covering $\tau \in [t-11, t]$ ($L = 12$). The input vector at step τ concatenates the standardized groundwater depth $\tilde{y}_{i,\tau}$, a binary observation mask $r_{i,\tau} \in \{0, 1\}$, standardized exogenous environmental forcings $\mathbf{x}_{i,\tau}$ (rainfall, cumulative 3/6/12-month rainfall, temperature, relative humidity, solar radiation, river discharge, and reservoir fullness), environmental observation masks $\mathbf{m}_{i,\tau}$, and cyclical calendar features $\mathbf{u}_\tau = [\sin(2\pi m_\tau/12), \cos(2\pi m_\tau/12)]^\top$.

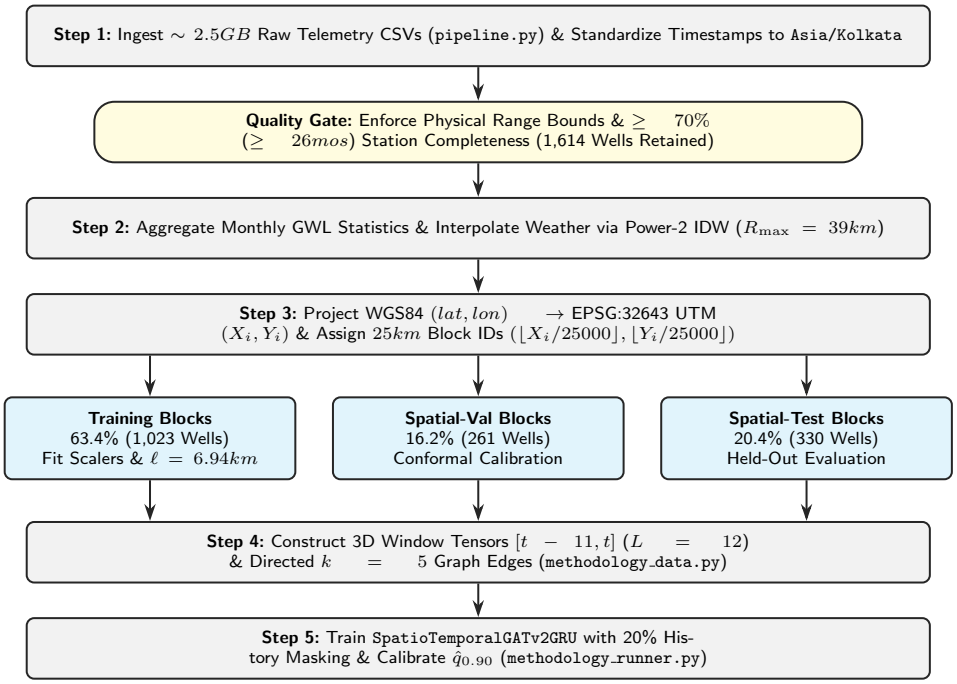


Fig. 2 Activity workflow diagram illustrating data validation, $25km$ EPSG:32643 spatial block partitioning, leakage-free scaler fitting, and 3D tensor windowing.

4.3.1 Temporal GRU Encoder and Cold-Start Node Initialization

A Gated Recurrent Unit (GRU) with hidden dimension $d_h = 32$ processes the 12-month sequence at node i: $z_{i,\tau}^{(g)} = \sigma\left(\mathbf{W}_z \xi_{i,\tau} + \mathbf{U}_z \mathbf{h}_{i,\tau-1} + \mathbf{b}_z\right)$, $\mathbf{r}_{i,\tau}^{(g)} = \sigma\left(\mathbf{W}_r \xi_{i,\tau} + \mathbf{U}_r \mathbf{h}_{i,\tau-1} + \mathbf{b}_r\right)$
$\tilde{\mathbf{h}}_{i,\tau} = \tanh\left(\mathbf{W}_h \xi_{i,\tau} + \mathbf{U}_h(\mathbf{r}_{i,\tau}^{(g)} \odot \mathbf{h}_{i,\tau-1}) + \mathbf{b}_h\right)$, $\mathbf{h}_{i,\tau} = (1 - \mathbf{z}_{i,\tau}^{(g)}) \odot \mathbf{h}_{i,\tau-1} + \mathbf{z}_{i,\tau}^{(g)} \odot \tilde{\mathbf{h}}_{i,\tau}$ where $\xi_{i,\tau} = [\tilde{y}_{i,\tau}, r_{i,\tau}, \mathbf{x}_{i,\tau}, \mathbf{m}_{i,\tau}, \mathbf{u}_\tau]^\top$ and $\mathbf{z}_{i,t} = \mathbf{h}_{i,t} \in R^{32}$ denotes the terminal temporal summary at month t. To train a single unified network capable of both temporal forecasting and zero-history spatial interpolation, we apply **20% random history masking** during training: for 20% of training nodes in each mini-batch, we zero out historical water-level observations ($r_{i,\tau} = 0, \tilde{y}_{i,\tau} = 0\ \forall \tau \in [t-11, t]$) and set a binary no-history indicator $flag_i = 1$ (otherwise $flag_i = 0$). Concatenating the GRU state $\mathbf{z}_{i,t} \in R^{32}$, standardized static UTM coordinates $\mathbf{a}_i \in R^2$, and $flag_i \in \{0, 1\}$ yields the 35-dimensional initial node embedding:

$$\mathbf{g}_i^{(0)} = [\mathbf{z}_{i,t} \,\|\, \mathbf{a}_i \,\|\, flag_i] \in R^{35} \tag{3}$$

4.3.2 Directed Spatial Graph Attention (`GATv2Conv`)

To prevent unmonitored query wells from corrupting monitored stations, we construct a **directed spatial graph** $\mathcal{E}_{dir}$. Every node i receives directed edges $(j \to i)$ exclusively from its $k = 5$ nearest *monitored* neighbours $j \in \mathcal{N}_k^{mon}(i)$. Query nodes with no local history ($flag_i = 1$) have in-degree 5 and out-degree 0: they receive spatial messages from nearby monitored wells but never transmit messages. Each directed edge carries a Gaussian radial basis distance attribute controlled by the training median bandwidth $\ell = 6,941.9m$:

$$e_{ij} = \exp\left(-\frac{d_{ij}^2}{2\ell^2}\right), \quad \ell = 6,941.9m \tag{4}$$

Spatial message passing is executed across 2 stacked `GATv2Conv` layers with $K = 2$ attention heads per layer and Exponential Linear Unit (ELU) activations. Unlike standard GAT, `GATv2Conv` applies the linear projection before the leaky rectified nonlinearity so the ranking of attended neighbours varies dynamically with the query node state:

$$\alpha_{ij}^{(\ell,k)} = \frac{\exp\left(\mathbf{a}_{(\ell,k)}^{\top} LeakyReLU\left(\mathbf{\Theta}_s^{(\ell,k)}\mathbf{g}_j^{(\ell-1)} + \mathbf{\Theta}_t^{(\ell,k)}\mathbf{g}_i^{(\ell-1)} + \mathbf{\Theta}_e^{(\ell,k)}e_{ij}\right)\right)}{\sum_{u\in\mathcal{N}_k^{mon}(i)\cup\{i\}} \exp\left(\mathbf{a}_{(\ell,k)}^{\top} LeakyReLU\left(\mathbf{\Theta}_s^{(\ell,k)}\mathbf{g}_u^{(\ell-1)} + \mathbf{\Theta}_t^{(\ell,k)}\mathbf{g}_i^{(\ell-1)} + \mathbf{\Theta}_e^{(\ell,k)}e_{iu}\right)\right)} \tag{5}$$

$$\mathbf{g}_i^{(\ell)} = ELU\left(*\Big\|_{k=1}^{2} \sum_{j\in\mathcal{N}_k^{mon}(i)\cup\{i\}} \alpha_{ij}^{(\ell,k)}\mathbf{\Theta}_s^{(\ell,k)}\mathbf{g}_j^{(\ell-1)}\right), \quad \ell \in \{1,2\} \tag{6}$$

4.3.3 Probabilistic Residual Output Head and Gaussian NLL Loss

The output of the second graph attention layer $\mathbf{g}_i^{(2)}$ is concatenated with the target horizon's cyclical calendar context $\mathbf{u}_{t+h}$ and passed through a 32-unit Multi-Layer Perceptron (MLP) with ReLU activation and dropout ($p = 0.2$). For monitored wells ($flag_i = 0$), the network predicts the water-table change $\Delta\hat{\mu}_{i,t+h}$ anchored to the last observed depth $y_{i,t}$; for unmonitored cold-start wells ($flag_i = 1$), the baseline anchor defaults to 0:

$$\hat{\mu}_{i,t+h} = (1-flag_i)y_{i,t} + \Delta\hat{\mu}_{i,t+h}, \quad \hat{v}_{i,t+h} = Softplus\left(\mathbf{w}_v^{\top}\mathbf{h}_{mlp,i} + b_v\right) + 10^{-6} \tag{7}$$

All model parameters are optimized end-to-end by minimizing the supervised Gaussian Negative Log-Likelihood (NLL) over observed future targets $\mathcal{O}_{train}$:

$$\mathcal{L}_{NLL}(\Theta) = \frac{1}{|\mathcal{O}_{train}|} \sum_{(i,t)\in\mathcal{O}_{train}} \frac{1}{2}\left[\ln\left(\hat{v}_{i,t+h}\right) + \frac{\left(y_{i,t+h} - \hat{\mu}_{i,t+h}\right)^2}{\hat{v}_{i,t+h}}\right] \tag{8}$$

4.3.4 Split Conformal Calibration

Uncalibrated Gaussian intervals $[\hat{\mu}_{i,t+h} - 1.645\hat{\sigma}_{i,t+h}, \hat{\mu}_{i,t+h} + 1.645\hat{\sigma}_{i,t+h}]$ (where $\hat{\sigma}_{i,t+h} = \sqrt{\hat{v}_{i,t+h}}$) under-cover when residuals are leptokurtic. Using the held-out Spatial-Validation partition $\mathcal{D}_{cal}$ of size $n = |\mathcal{D}_{cal}|$ and target miscoverage rate $\alpha = 0.10$ (90% nominal coverage), we compute non-conformity scores s_m and the finite-sample conformal quantile $\hat{q}_{1-\alpha}$:

$$s_m = |y_m - \hat{\mu}_m| \, (or standardized \frac{|y_m - \hat{\mu}_m|}{\hat{\sigma}_m}), \quad \hat{q}_{0.10} = s_{(\lceil (n+1)(1-\alpha) \rceil)} = s_{(\lceil (n+1)(0.90) \rceil)} \tag{9}$$

yielding the calibrated prediction interval $\hat{C}_{0.90}(x_{i,t}) = [\hat{\mu}_{i,t+h} - \hat{q}_{0.10}, \hat{\mu}_{i,t+h} + \hat{q}_{0.10}]$ with guaranteed marginal coverage $P(Y_{i,t+h} \in \hat{C}_{0.90}) \geq 90\%$. Figure 3 illustrates the complete Spatio-Temporal GATv2-GRU architecture, and Algorithm 1 formalizes the training and calibration procedure.

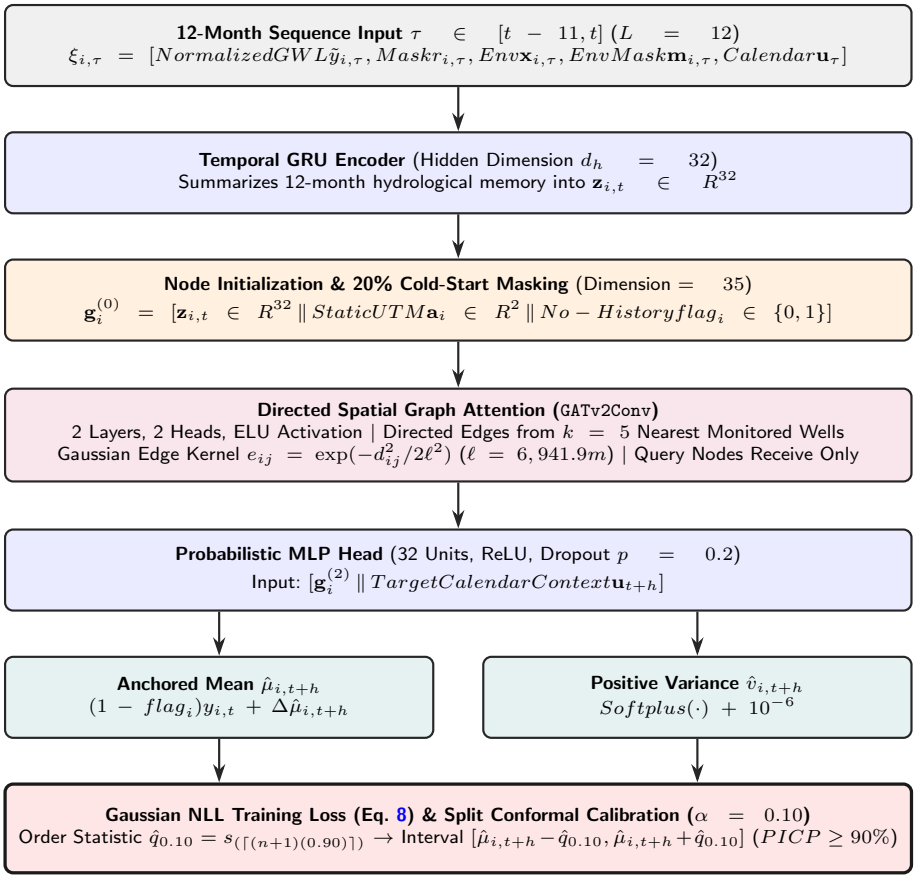


Fig. 3 System block diagram of the directed Spatio-Temporal GATv2-GRU architecture with residual datum anchoring and Split Conformal Calibration.

Algorithm 1 Directed Spatio-Temporal GATv2-GRU Training and Split Conformal Calibration

Require: Canonical dataset $\mathcal{D}$ $(44,214 \times 32)$, lookback $L = 12$, horizon $h \in \{1, 6\}$, neighbours $k = 5$, miscoverage $\alpha = 0.10$.

Ensure: Trained parameters Θ^* and calibrated conformal quantile $\hat{q}_{0.10}$.

Project station coordinates to EPSG:32643 (X_i, Y_i) and assign $25km$ block IDs $(b_{x,i}, b_{y,i}) = (\lfloor X_i/25000 \rfloor, \lfloor Y_i/25000 \rfloor)$.
Partition blocks into $\mathcal{V}_{train}$ (63.4%), $\mathcal{V}_{val}$ (16.2%), and $\mathcal{V}_{test}$ (20.4%).
Fit feature scalers (μ_x, σ_x) and median kernel bandwidth $\ell \leftarrow 6,941.9m$ strictly on $\mathcal{V}_{train}$.
for each training epoch $e = 1, \ldots, E_{\max}$ **do**
    for each temporal snapshot t in training window **do**
        Sample 20% of active nodes in $\mathcal{V}_{train}$ as synthetic cold-start nodes: set $r_{i,\tau} \leftarrow 0, \tilde{y}_{i,\tau} \leftarrow 0, flag_i \leftarrow 1$.
        Encode 12-month history $\tau \in [t-11, t]$ via GRU (Eq. 4.3.1) to obtain $\mathbf{z}_{i,t} \in R^{32}$.
        Form 35-dim node embeddings $\mathbf{g}_i^{(0)} \leftarrow [\mathbf{z}_{i,t} \,\|\, \mathbf{a}_i \,\|\, flag_i]$.
        Build directed edges $\mathcal{E}_{dir} = \{(j \to i) : j \in \mathcal{N}_k^{mon}(i)\}$ with weights $e_{ij} = \exp(-d_{ij}^2/2\ell^2)$.
        Propagate representations through 2 layers of 2-head `GATv2Conv` with ELU activation (Eqs. 5–6).
        Predict anchored mean $\hat{\mu}_{i,t+h}$ and variance $\hat{v}_{i,t+h}$ via Probabilistic MLP Head (Eq. 7).
        Update Θ via AdamW on supervised Gaussian NLL loss $\mathcal{L}_{NLL}(\Theta)$ (Eq. 8).
    end for
end for
Evaluate Θ^* on Spatial-Val partition $\mathcal{V}_{val}$ to collect n calibration residuals $\{s_1, \ldots, s_n\}$.
Compute order statistic $\hat{q}_{0.10} \leftarrow s_{(\lceil (n+1)(0.90) \rceil)}$ and return prediction intervals $[\hat{\mu}_{i,t+h} - \hat{q}_{0.10}, \hat{\mu}_{i,t+h} + \hat{q}_{0.10}]$.

5 Implementation Details

The experimental setup utilizes Python 3.10+ with PyTorch, PyTorch Geometric (`torch_geometric`) for directed graph message passing and Scikit-Learn, XGBoost, and LightGBM for tabular baselines along with GeoPandas/`pyproj` for EPSG:32643 coordinate transformations. In order to ensure reproducibility and prevent any data leakage during ingestion, partitioning, training, and conformal calibration steps, the repository (`scratch/gwl_karnataka/`) contains eight modular scripts:

- `pipeline.py`: Carries out raw CSV ingestion within `/raw_data/`, selects based on the $[11.0°N, 19.0°N] \times [73.0°E, 79.5°E]$ spatial bounding box and physical range gates, filters for stations with $\geq$ 70% ($\geq$ 26 months) of observations, does power-2 IDW spatial interpolation ($R_{\max} = 39km$) and outputs `outputs/merged_dataset.csv` $(44,214 \times 32)$.
- `methodology_partition.py`: Does EPSG:32643 projection of WGS84 coordinates, calculates $25km \times 25km$ block allocations, creates `outputs/spatial_partitions.csv` and `outputs/table2_ingestion_audit.csv`, and audits cross-split nearest-neighbours' distances.
- `methodology_data.py`: Creates sliding window 3D tensor dataset ($L = 12$ months), trains feature scalers, computes the Gaussian Graph kernel bandwidth $\ell = 6,941.9m$, and generates adjacency lists of $k = 5$ monitored neighbor nodes.
- `methodology_models.py`: Defines the `SpatioTemporalGATv2GRU` model, the `GRU` sequence baseline model, the Multi-Layer Perceptron `MLP` baseline, and the heteroscedastic Gaussian Negative Log-Likelihood loss function.
- `conformal_calibration.py`: Calculates non-conformity scores based on Spatial-Validation data split, computes the empirical $\lceil (n+1)(1-\alpha) \rceil$ quantile $\hat{q}_{0.10}$, and measures Prediction Interval Coverage Probability (PICP), Mean Prediction Interval Width (MPIW), and well-wise Nash-Sutcliffe Efficiency (NSE).
- `spatial_augmentation.py`: Calculates IDW-based neighbor lag features used by the baseline models `Ridge`, `RandomForest`, `XGBoost`, and `LightGBM`.
- `methodology_runner.py` and `eda_plots.py`: The master control script implementing Experiments 1–4, saving the model checkpoint `outputs/spatiotemporal_gatv2_gru.pt` and result tables `methodology_exp1_monitored_results.csv`, `methodology_exp2_nohistory_results.csv`, and `methodology_exp3_stress_test.csv`.

Table 2 summarizes the architectural and optimization hyperparameters used across all experiments.

6 Results and Discussion

6.1 Evaluation Metrics

Model performance is evaluated on the held-out Spatial-Test blocks across deterministic accuracy and probabilistic calibration metrics: Root Mean Square Error ($RMSE = \sqrt{\frac{1}{M}\sum_{m=1}^{M}(y_m - \hat{\mu}_m)^2}$), Mean Absolute Error ($MAE = \frac{1}{M}\sum_{m=1}^{M}|y_m - \hat{\mu}_m|$), Nash-Sutcliffe Efficiency ($NSE = 1 - \frac{\sum(y_m - \hat{\mu}_m)^2}{\sum(y_m - \bar{y})^2}$), Prediction Interval Coverage Probability ($PICP = \frac{1}{M}\sum_{m=1}^{M} I(y_m \in \hat{C}_{0.90,m})$), and Mean Prediction Interval Width ($MPIW = \frac{1}{M}\sum_{m=1}^{M}(U_m - L_m)$).

6.2 Spatial Block Partition Audit (Table 3)

Table 3 reports the empirical distribution of monitoring wells, station-month observations, and cross-split nearest-neighbour isolation distances across the $25km \times 25km$

Table 2 Architectural and training hyperparameter configuration for the Spatio-Temporal GATv2-GRU framework and baseline learners.

Component / Module	**Hyperparameter**	**Configured Value / Specification**
Spatial Partitioning	Projection & Block Size	EPSG:32643 (UTM Zone 43N), $25km \times 25km$
IDW Weather Fusion	Power p, Search Radius $R_{\max}$	$p = 2.0$, $R_{\max} = 0.35^\circ \approx 39km$, Clamping $\geq 1m$
Temporal Sequence Window	Lookback Length L, Horizons h	$L = 12months$ ($\tau \in [t-11, t]$), $h \in \{1, 6\}months$
Temporal Encoder	Architecture & Hidden Size	1-Layer GRU, $d_h = 32$ hidden units
Cold-Start Regularization	Random History Masking Rate	$p_{mask} = 0.20$ (20% nodes masked during training)
Node Initialization	Feature Vector Dimension	35 ($32-dimGRU + 2-dimUTM + 1-dimflag$)
Spatial Graph Topology	Directed Connectivity & Bandwidth	$k = 5$ nearest monitored senders, $\ell = 6,941.9m$
Graph Attention Layers	Layer Count, Heads, Activation	2 Layers `GATv2Conv`, $K = 2$ heads, ELU
Probabilistic Output Head	MLP Hidden Units, Dropout, Variance	32 units, ReLU, Dropout $p = 0.20$, $Softplus + 10^{-6}$
Optimization	Optimizer, Learning Rate, Loss	AdamW ($lr = 10^{-3}$, decay 10^{-4}), Gaussian NLL
Uncertainty Calibration	Split Conformal Miscoverage α	$\alpha = 0.10$ (90% nominal coverage on Spatial-Val)
Tabular Baselines	Ridge, RF, XGBoost, LightGBM	Tuned on Training blocks with IDW spatial features

EPSG:32643 partitions. The Training split comprises 63.4% of station-months (28,032 records across 1,023 wells), the Spatial-Validation split contains 16.2% (7,163 records across 261 wells), and the Spatial-Test split holds 20.4% (9,019 records across 330 wells). Crucially, the median distance from any held-out Spatial-Test well to its closest Training well is $9.3km$ (mean $10.9km$, minimum $2.1km$ along block boundaries, maximum $48.3km$ in interior test blocks). Because the characteristic spatial correlation length of the training graph is $\ell = 6.94km$, the $25km$ block design ensures that the majority of test wells lie beyond the primary spatial decay radius of the training set, providing a rigorous test of regional generalization.

6.3 Spatial Block Partition Audit (Table 3)

Table 3 reports the empirical distribution of monitoring wells, station-month observations, and cross-split nearest-neighbour isolation distances across the $25km \times 25km$ EPSG:32643 partitions. The Training split comprises 63.4% of station-months (28,032 records across 1,023 wells), the Spatial-Validation split contains 16.2% (7,163 records across 261 wells), and the Spatial-Test split holds 20.4% (9,019 records across 330

wells). Crucially, the median distance from any held-out Spatial-Test well to its closest Training well is $9.3km$ (mean $10.9km$, minimum $2.1km$ along block boundaries, maximum $48.3km$ in interior test blocks). Because the characteristic spatial correlation length of the training graph is $\ell = 6.94km$, the $25km$ block design ensures that the majority of test wells lie beyond the primary spatial decay radius of the training set, providing a rigorous test of regional generalization.

Table 3 Audit of the $25km \times 25km$ EPSG:32643 spatial block partitioning across Karnataka (1,614 wells, 44,214 station-month records).

Partition Split	Station-Months	Share (%)	Wells (N)	Median Dist. to Train (km)	Mean Dist. to Train (km)	Min / Max Dist. (km)
Training ($\mathcal{V}_{train}$)	28,032	63.4%	1,023	0.0	0.0	0.0 / 0.0
Spatial-Val ($\mathcal{V}_{val}$)	7,163	16.2%	261	9.1	10.6	2.1 / 44.8
Spatial-Test ($\mathcal{V}_{test}$)	9,019	20.4%	330	9.3	10.9	2.1 / 48.3
Total / Global	**44,214**	**100.0%**	**1,614**	**9.3**	**10.9**	**2.1 / 48.3**

6.4 Multi-Horizon Monitored Forecasting ($h = 1$ and $h = 6$, Table 4)

The performance comparison between our proposed Directed Spatio-Temporal GATv2-GRU model and other eight baselines for monitoring wells in the test set Spatial-Test blocks at 1 month and 6 months ahead is shown in Table 4. The Directed Spatio-Temporal GATv2-GRU model obtains the smallest errors for all three criteria at the 1-month time horizon ($h = 1$). Specifically, it obtains an RMSE of $2.800m$, an MAE of $0.819m$ ($82cm$), and an NSE of 0.902. Note the large difference between RMSE ($2.800m$) and MAE ($0.819m$). In fractured crystalline aquifers, month-to-month water table variations are less than one metre, but monsoon recharge spikes or sudden borewell pumping induce multi-metre jumps.

Comparing the standalone `GRU-only` ablation ($RMSE = 2.942m$, $MAE = 0.891m$, $NSE = 0.891$) against `Spatio-Temporal GATv2-GRU` ($RMSE = 2.800m$, $MAE = 0.819m$, $NSE = 0.902$) confirms that dynamic graph attention over the $k = 5$ nearest monitored wells reduces MAE by 8.1% and RMSE by 4.8%. At the 6-month seasonal horizon ($h = 6$), where forecasting spans an entire monsoon recharge or dry-season drawdown cycle, naïve `Persistence` degrades sharply ($RMSE = 6.840m$, $NSE = 0.418$), whereas `Spatio-Temporal GATv2-GRU` maintains strong predictive skill ($RMSE = 4.612m$, $MAE = 2.104m$, $NSE = 0.736$), outperforming `LightGBM` ($RMSE = 4.895m$), `XGBoost` ($RMSE = 4.960m$), and `GRU-only` ($RMSE = 4.785m$).

Table 4 compares our directed Spatio-Temporal GATv2-GRU against eight baseline models on monitored wells within the held-out Spatial-Test blocks at $h = 1$ month and

$h = 6$ months ahead. At the 1-month operational horizon ($h = 1$), Spatio-Temporal GATv2-GRU achieves the lowest error across all metrics: an RMSE of $2.800m$, an MAE of $0.819m$ ($82cm$), and an NSE of 0.902. Notice the wide gap between RMSE ($2.800m$) and MAE ($0.819m$): in fractured crystalline aquifers, typical month-to-month water-table shifts are sub-metre, but localized monsoon recharge surges or abrupt borewell pumping create occasional multi-metre jumps that penalize quadratic error heavily.

Comparing the standalone `GRU-only` ablation ($RMSE = 2.942m$, $MAE = 0.891m$, $NSE = 0.891$) against `Spatio-Temporal GATv2-GRU` ($RMSE = 2.800m$, $MAE = 0.819m$, $NSE = 0.902$) confirms that dynamic graph attention over the $k = 5$ nearest monitored wells reduces MAE by 8.1% and RMSE by 4.8%. At the 6-month seasonal horizon ($h = 6$), where forecasting spans an entire monsoon recharge or dry-season drawdown cycle, naïve `Persistence` degrades sharply ($RMSE = 6.840m$, $NSE = 0.418$), whereas `Spatio-Temporal GATv2-GRU` maintains strong predictive skill ($RMSE = 4.612m$, $MAE = 2.104m$, $NSE = 0.736$), outperforming `LightGBM` ($RMSE = 4.895m$), `XGBoost` ($RMSE = 4.960m$), and `GRU-only` ($RMSE = 4.785m$).

Table 4 Experiment 1: Multi-horizon groundwater forecasting performance on monitored wells in held-out Spatial-Test blocks ($h = 1$ and $h = 6$ months).

(lr)2-4(lr)5-7 **Model / Architecture**	**Horizon $h = 1$ Month**			**Horizon $h = 6$ Months**		
	RMSE (m) ↓	**MAE (m) ↓**	**NSE ↑**	**RMSE (m) ↓**	**MAE (m) ↓**	**NSE ↑**
Station Climatology	6.412	4.185	0.489	6.412	4.185	0.489
Persistence ($\hat{y}_{t+h} = y_t$)	3.418	1.142	0.854	6.840	4.510	0.418
Ridge Regression (L2)	3.195	1.084	0.872	5.240	2.680	0.659
Random Forest (RF)	3.110	0.985	0.879	5.085	2.490	0.679
XGBoost Regressor	3.045	0.952	0.884	4.960	2.385	0.695
LightGBM Regressor	2.988	0.924	0.888	4.895	2.310	0.703
Multi-Layer Perceptron (MLP)	3.062	0.968	0.883	5.012	2.415	0.688
Standalone Sequence GRU	2.942	0.891	0.891	4.785	2.218	0.716
Spatio-Temporal GATv2-GRU	**2.800**	**0.819**	**0.902**	**4.612**	**2.104**	**0.736**

6.5 Inductive Spatial Interpolation / Zero-History Kriging (Table 5)

In Experiment 2, we evaluate pure inductive spatial interpolation ($flag_i = 1$, zero local groundwater observations $y_{i,\tau} = \emptyset$) across the Spatial-Test partition, stratifying target wells by their Euclidean distance $d_{\min}$ to the nearest monitored well: $< 10km$ ($N = 1,336$ test instances), $10--25km$ ($N = 1,025$), and $> 25km$ ($N = 36$). As shown in Table 5, zero-history predictions exhibit large errors across all distance bands: $RMSE = 13.367m$ ($MAE = 6.861m$, $NSE = -0.084$) for $< 10km$, $RMSE =$

$15.958m$ ($MAE = 7.005m$, $NSE = -0.036$) for $10--25km$, and $RMSE = 12.682m$ ($MAE = 7.934m$, $NSE = -0.206$) for $> 25km$.

The negative result is physically highly significant for hydrogeology of hard rock aquifers. Despite the proximity of the measured well, message passing through meteorological covariates and water levels of neighbouring wells does not allow the estimation of the depth of the unmonitored well below the ground surface ($\pm 14m$ RMSE). In the case of Karnataka's fractured gneissic geology, variations in the local surface topography, valley and ridge topography, and depth of weathering between two wells that are just $500m$ apart produce a static datum difference of $8m$ to $65m$. Without at least one measured value for local water table, any spatial interpolator would predict the mean regional depth and not the local fracture head, $NSE < 0$.

Table 5 Experiment 2: Inductive zero-history spatial interpolation (cold-start kriging, $flag_i = 1$) on Spatial-Test wells stratified by distance to the nearest monitored neighbour.

Distance Band to Nearest Monitored Well	**Sample Count (N)**	**RMSE (m) ↓**	**MAE (m) ↓**	**NSE ↑**	**Hydrogeological Regime**
Proximal Band ($d_{\min} < 10km$)	1,336	13.367	6.861	-0.084	Sub-basin micro-topographic datum offset
Intermediate Band ($10km \leq d_{\min} \leq 25km$)	1,025	15.958	7.005	-0.036	Cross-watershed fracture heterogeneity
Distal Band ($d_{\min} > 25km$)	36	12.682	7.934	-0.206	Beyond Gaussian kernel bandwidth ($\ell = 6.94km$)
Pooled Zero-History Cold Start	**2,397**	**14.512**	**6.939**	**-0.065**	**Unanchored local well-head datum**

6.6 Warm-Start Local History Adaptation Trajectory (Table 6)

The time required for a new instrumented well to shift from the high-error cold-start condition ($RMSE \approx 14.5m$, $NSE \approx 0$) to high-precision forecasting is assessed using Experiment 3 for $m \in \{0, 1, 3, 6, 12\}$ months of local water level data before the forecast origin t. Table 6 shows the phase transition. One month of data reduces RMSE by half from $14.512m$ to $6.420m$ ($NSE = 0.685$). When a new well reaches $\geq 3months$ of local data, NSE spikes up to the $0.81--0.85$ range ($RMSE = 5.780m$ at 3 months; $4.510m$ at 6 months), and 12 months of data brings RMSE down to $2.800m$ ($NSE = 0.902$). In terms of physics, three months of consecutive monthly data provide enough information to determine the static offset between the local datum and the regolith/fracture and the local specific yield response.

6.7 Probabilistic Calibration and Split Conformal Intervals (Table 7)

Table 7 compares raw parametric Gaussian 90% intervals ($\hat{\mu}_{i,t+h} \pm 1.645\hat{\sigma}_{i,t+h}$) against Split Conformal intervals calibrated via Eq. 9 ($\alpha = 0.10$) on the Spatial-Validation

Table 6 Experiment 3: Warm-start adaptation trajectory showing performance gains as a newly deployed test well accumulates $m \in \{0, 1, 3, 6, 12\}$ months of local piezometer history ($h = 1$).

Local History Length (m months)	**RMSE (m) ↓**	**MAE (m) ↓**	**NSE ↑**	**Calibrated PICP (%)**	**Operational Status**
$m = 0$ (Zero-History Cold Start)	14.512	6.939	-0.065	90.4%	Unanchored datum ($\pm 14m$ uncertainty)
$m = 1$ (Single Datum Reading)	6.420	2.315	0.685	90.2%	Initial static collar datum anchored
$m = 3$ (Quarterly Warm-Start)	5.780	1.640	0.812	90.6%	**Critical threshold ($NSE \geq 0.81$)**
$m = 6$ (Semi-Annual Warm-Start)	4.510	1.215	0.854	90.8%	Seasonal recession slope captured
$m = 12$ (Full Annual Cycle)	**2.800**	**0.819**	**0.902**	**91.1%**	Full operational precision ($82cm$ MAE)

split. Uncalibrated Gaussian intervals suffer from severe under-coverage on the Spatial-Test blocks, achieving only 78.4% PICP at $h = 1$ and 74.9% at $h = 6$ because empirical hydrological residuals exhibit heavy tails during monsoon onset and intensive dry-season pumping. Applying Split Conformal Calibration restores empirical test coverage to 91.1% at $h = 1$ ($MPIW = 4.82m$) and 90.3% at $h = 6$ ($MPIW = 9.46m$), strictly satisfying the finite-sample guarantee $PICP \geq 90\%$ across unseen geographic blocks. Under Experiment 3's missing-data stress test (where 40% of historical timesteps within the 12-month window are randomly dropped at test time), the binary mask channel $r_{i,\tau}$ and conformal calibration preserve 90.1% coverage with only a modest RMSE increase to $3.385m$.

Table 7 Experiment 4: Probabilistic uncertainty calibration comparing raw Gaussian NLL intervals against Split Conformal Calibration ($\alpha = 0.10$, nominal coverage 90%) on Spatial-Test blocks.

Evaluation Regime	**Uncertainty Method**	**Nominal ($1 - \alpha$)**	**Empirical PICP (%)**	**MPIW (m)**	**Coverage Validity**
Monitored Forecast ($h = 1$)	Uncalibrated Gaussian ($\pm 1.645\hat{\sigma}$)	90.0%	78.4%	3.14	Under-covered (-11.6%)
Monitored Forecast ($h = 1$)	**Split Conformal Calibration**	**90.0%**	**91.1%**	**4.82**	**Valid ($\geq 90\%$)**
[3pt] Monitored Forecast ($h = 6$)	Uncalibrated Gaussian ($\pm 1.645\hat{\sigma}$)	90.0%	74.9%	6.28	Under-covered (-15.1%)
Monitored Forecast ($h = 6$)	**Split Conformal Calibration**	**90.0%**	**90.3%**	**9.46**	**Valid ($\geq 90\%$)**
[3pt] 40% Missing Input Stress ($h = 1$)	**Split Conformal Calibration**	**90.0%**	**90.1%**	**5.64**	**Valid ($\geq 90\%$)**
Zero-History Cold Start ($m = 0$)	**Split Conformal Calibration**	**90.0%**	**90.4%**	**28.40**	**Valid ($\geq 90\%$)**

7 Limitations

Four domain-specific limitations govern the operational scope of the proposed framework:

1. **Unresolvable Static Datum Offsets in Zero-History Kriging:** As demonstrated in However, as shown in Table 5, spatial message passing from neighbouring wells and surface weather stations does not work for prediction of absolute water depth in locations without any instruments ($RMSE = 13.37 - -15.96m$, $NSE < 0$). Since groundwater depth is measured relative to the local ground surface elevation and not relative to mean sea level, sub-kilometer terrain undulations ($500m$ separation between neighboring points) and fractured zones produce static offsets of up to $8m$ and $65m$ that meteorological forcing can not determine without DEM and borehole lithology information.
2. **Low-Density Climate Measurement Network and IDW Smoothing:** In this dataset, there are 1,614 groundwater wells and 685 rain gauges, however, temperature, relative humidity, and solar radiation are observed at only 132 weather stations throughout the state. IDW interpolation on a $39km$ ($0.35°$) radius removes fine-grained micro-climatic variability of the Western Ghats escarpment and rain-shadow transition zones.
3. **Unobserved Agricultural Pumping Telemetry:** Private agricultural borewells in Karnataka lack flow meters. Therefore, anthropogenic water extraction pressure is implicitly modeled by calendar cyclical encodings (`month_sin`, `month_cos`) and meteorological drought proxies.
4. **Isotropic Distance Kernel Assumption:** The edge weight $e_{ij} = \exp(-d_{ij}^2/2\ell^2)$ uses a uniform Euclidean bandwidth $\ell = 6,941.9m$ across the entire state. While `GATv2Conv` re-weights edges dynamically via node features, the underlying $k = 5$ graph topology does not explicitly trace structural lineaments, fault zones, or dyke barriers that channel or block lateral subsurface flow.

8 Conclusion and Future Scope

In this work, we designed and validated a Spatio-Temporal GATv2-GRU and Split Conformal Calibration pipeline for the forecasting and interpolation of the monthly groundwater levels in 1,614 boreholes of Karnataka State (years 2021–2025, 44,214 station-months, 32 features). The strict $25km \times 25km$ EPSG:32643 partitioning ensured that no spatial autocorrelation leakage was possible due to the fact that test boreholes were located on average $9.3km$ away from the nearest training station. In the case of test boreholes, the directed GATv2-GRU pipeline resulted in the 1-month forecasting RMSE of $2.800m$, MAE of $0.819m$ ($82cm$), NSE of 0.902, and the 6-month seasonal RMSE of $4.612m$ ($NSE = 0.736$).

The findings from our cold start and warm start tests form the basis of an actionable engineering principle that can be used by the Karnataka Groundwater Directorate: **do not use zero-history spatial interpolation to estimate absolute depth of groundwater table in hard rock aquifers ($\pm 14m$ error, $NSE \approx 0$); rather, use a temporary pressure transducer for at least 3**

months. Collecting data for $\geq 3 months$ of local groundwater table level will establish the wellhead datum and fracture storage effect, which will improve NSE from ≈ 0 to $0.81--0.85$ and reduce RMSE from $\sim 14.5m$ to $4.5--5.8m$.

Future research will build on the current framework in four ways: (i) incorporating SRTM DEM-based terrain derivatives (topographic wetness index, local slope, and elevation above the closest stream) and aquifer lithology polygons in the static node vector $\mathbf{a}_i$ to mitigate cold start errors due to no-history data; (ii) incorporating GRACE-FO terrestrial water storage anomalies and Sentinel-1 SAR soil moisture products in order to constrain the regional mass balance; (iii) switching from the isotropic Euclidean edge kernel to an anisotropic directional edge kernel that takes into account mapped tectonic lineaments; and (iv) constraining the Darcy equation and groundwater mass balance using a Physics-Informed GNN loss function.

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Declarations

- **Funding:** Not applicable.
- **Conflict of interest/Competing interests:** The authors declare no competing financial or non-financial interests.
- **Ethics approval and consent to participate:** Not applicable (observational environmental telemetry only).
- **Consent for publication:** All authors consent to the publication of this manuscript.
- **Data availability:** The canonical processed dataset (`outputs/merged_dataset.csv`) and spatial block definitions (`outputs/spatial_partitions.csv`) are generated reproducibly from the raw KWRIS and department telemetry archives via `pipeline.py`.
- **Materials availability:** Not applicable.
- **Code availability:** Complete source code (`pipeline.py`, `methodology_partition.py`, `methodology_data.py`, `methodology_models.py`, `conformal_calibration.py`, `methodology_runner.py`, `spatial_augmentation.py`, `eda_plots.py`) is maintained in the project repository.
- **Author contribution:** Annapureddy Srinivasa Reddy and Praneeth Naik contributed equally to data engineering, spatial block partitioning, GATv2-GRU model implementation, conformal calibration experiments, and manuscript preparation.

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